VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH MERN

Project Planning

Online Complaint Registration System — MERN Stack Complaint Management Platform

Phase: Brainstorming & Ideation Phase

Document Version: 1.1

Date: 18th of june, 2026

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Classification: Project Documentation — For Academic Submission

Project Planning

1. Introduction

This document formalizes the project planning activities for the Online Complaint Registration System development initiative [cite: 711]. The project planning framework follows Agile Scrum methodology and includes Product Backlog creation, Sprint Planning, Story Point Estimation, Sprint Timeline, Velocity Tracking, Burndown Analysis, and Risk Assessment [cite: 712, 713].

The Online Complaint Registration System is a MERN Stack web application developed to streamline complaint registration, complaint tracking, service agent assignment, citizen communication, administrative governance, feedback collection, and analytics-driven monitoring. The project execution strategy was divided into three major development sprints [cite: 713]:

- **Sprint 1** – Foundation & Authentication
- **Sprint 2** – Complaint Management & Agent Operations
- **Sprint 3** – Real-Time Communication, Feedback & Analytics

2. Product Backlog & Sprint Schedule

Sprint	Epic [cite: 716]	USN [cite: 717]	User Story / Task [cite: 719]	Points [cite: 718]	Priority [cite: 718]
Sprint 1	Project Setup & DB [cite: 721]	USN-0 [cite: 722]	Initialize React frontend, Express backend, MongoDB Atlas connection, project folder structure, environment configuration, and GitHub repository setup [cite: 723].	3 [cite: 724]	**High** [cite: 725]
Sprint 1	User Auth [cite: 736]	USN-1 [cite: 737]	Citizen can register an account using name, email, and password and receive a JWT session [cite: 738, 742].	2 [cite: 739]	**High** [cite: 740]
Sprint 1	User Auth [cite: 749]	USN-2 [cite: 750]	User can log in securely using credentials and be granted a JWT in an HTTP-only cookie [cite: 751, 755].	2 [cite: 752]	**High** [cite: 753]
Sprint 1	Role Management	USN-3	Administrator can manage user roles and access permissions securely.	2	**High**
Sprint 2	Complaint Registration	USN-4 [cite: 786]	Citizen can submit complaints with issue details and category selection.	3 [cite: 788]	**High** [cite: 789]
Sprint 2	Complaint Tracking	USN-5 [cite: 798]	Citizen can monitor complaint progress through status tracking and timeline updates.	3 [cite: 810]	**High** [cite: 811]
Sprint 2	Agent Assignment	USN-6 [cite: 816]	Administrator can assign complaints to service agents with status mapping.	3	**High** [cite: 815]
Sprint 2	Agent Workspace	USN-7 [cite: 829]	Service agents can view assigned complaints and update tracking statuses [cite: 825, 830].	3	**High**
Sprint 3	Real-Time Communication	USN-8 [cite: 841]	Citizens and service agents can communicate using Socket.IO chat functionality [cite: 837].	5	**High**
Sprint 3	Notification System	USN-9 [cite: 859]	Users receive notifications for assignments, status changes, and complaint resolutions.	3 [cite: 856]	**Medium** [cite: 857]
Sprint 3	Feedback System	USN-10	Citizens can submit ratings and feedback reviews after complaint resolution.	2	**Medium**
Sprint 3	Analytics Dashboard [cite: 858]	USN-11	Administrator can view complaint analytics, metrics, and performance reports.	3	**Medium**

3. Sprint Timeline — Gantt Chart

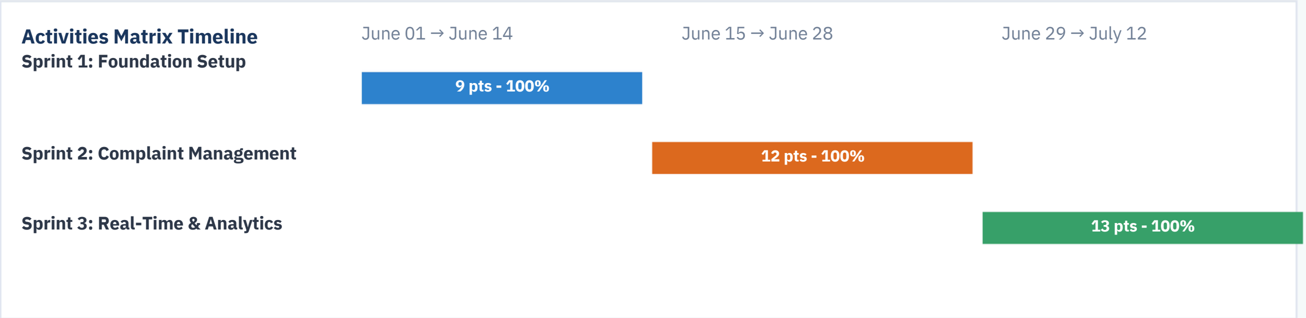

FIGURE 3.1 — GANTT CHART: SPRINT DELIVERY TIMELINE (3 SPRINTS × 2 WEEKS) [CITE: 879]

4. Velocity & Project Tracker

Sprint [cite: 681]	Story Points [cite: 681]	Start Date [cite: 681]	End Date [cite: 681]	Completed [cite: 681]	Completion Rate [cite: 882]
Sprint 1	9	10 June 2026 [cite: 882]	13 June 2026 [cite: 882]	9 [cite: 882]	100% [cite: 882]
Sprint 2	12	13 June 2026 [cite: 882]	15 June 2026 [cite: 882]	12 [cite: 882]	100% [cite: 882]
Sprint 3	13	15 June 2026 [cite: 882]	17 July 2026 [cite: 882]	13 [cite: 882]	100% [cite: 882]
Total	**34**	--		**34** [cite: 882]	**100%** [cite: 882]

Velocity Analysis: [cite: 682]
Total Story Points Completed = 34. Average Velocity = 11.3 Story Points per Sprint [cite: 883]. Sprint Duration = 2 Weeks [cite: 883]. Project Completion Rate = 100% [cite: 882].

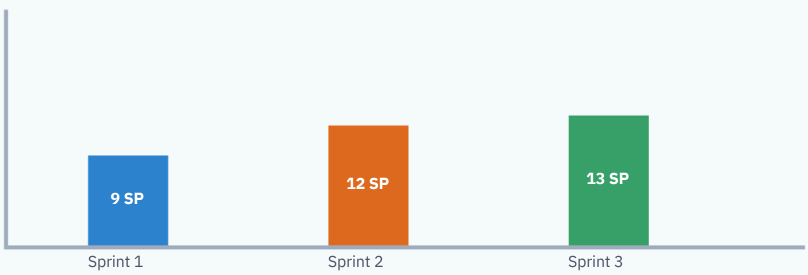

FIGURE 4.1 — TEAM VELOCITY TRACKING ACROSS DEVELOPMENT SPRINTS

5. Burndown Analysis

Aburndown chartrepresents work left to do versus time [cite: 687]. It provides precise visibility over iteration metrics [cite: 689].

6. Risk Register

Risk ID	Risk Description [cite: 884]	Probability [cite: 884]	Impact [cite: 884]	Mitigation Strategy [cite: 884]
R-01	MongoDB Atlas connectivity issues during deployment [cite: 885]	Medium [cite: 885]	High [cite: 885]	Configure connection retry logic, maintain Atlas monitoring and backup configurations [cite: 885].
R-02	JWT authentication vulnerabilities or token exposure [cite: 885]	Low [cite: 885]	Critical [cite: 885]	Store secrets safely in .env files (git-ignored); enforce token validation constraints [cite: 885].
R-03	Unauthorized access to complaint information	Low	High	Implement strict role-based access control (RBAC) rules and protected backend validation endpoints.
R-04	Socket.IO communication failures affecting chat functionality [cite: 885]	Medium [cite: 885]	Medium [cite: 885]	Implement reconnection orchestration handling and robust socket event schema validation [cite: 885].
R-05	High complaint volume affecting system performance	Medium	Medium	Utilize scalable MongoDB Atlas sharding profiles and optimize collection indexing structures.
R-06	Notification delivery failures	Low	Medium	Implement background task notification retry brokers with fallback tracing mechanisms.
R-07	Data inconsistency during complaint status updates	Low	High	Use isolated atomic transactional write operations alongside server-side logging layers.
R-08	Frontend deployment configuration issues [cite: 885]	Medium [cite: 885]	Medium [cite: 885]	Maintain precise configuration environment injection profiles across deployment pipelines [cite: 885].
R-09	User adoption challenges due to unfamiliarity with system	Medium	Medium	Provide streamlined, clear, material-driven layout guidelines and context-aware walkthrough hints.
R-10	Service downtime during maintenance windows	Low	High	Schedule deployment intervals during low-traffic slots; integrate blue-green zero-downtime hooks.

7. Conclusion

TheAgile Scrum-based planning framework provided a structured and iterative approach for developing the Online Complaint Registration System [cite: 887]. All 34 story points across the multiple epics were completed with a 100% sprint completion rate within the planned six-week timeline [cite: 888]. This methodology established a robust foundation for successful execution, operational testing, deployment, and future software scalability [cite: 888].

Additional Resources Required

- Agile execution tracker log sheets or Jira workspace sprint velocity export dashboards.
- Institutional review board verification signatures and deployment timeline authorization waivers.